

Regarding ModelMapper :

In order to automatically convert Entity <----> DTO , there are multiple 3rd party libraries.
eg : MapStruct , ModelMapper ...

Steps for using Model Mapper :

1.1 Add model mapper dependency in pom.xml

```
<!-- https://mvnrepository.com/artifact/org.modelmapper/modelmapper -->
<dependency>
  <groupId>org.modelmapper</groupId>
  <artifactId>modelmapper</artifactId>
  <version>3.0.0</version>
</dependency>
```

1.2 How to configure ModelMapper object as a spring bean (so that it can be injected as a dependency in other spring beans) ?

Ans : Either use xml <bean> tag OR use

@Bean annotated method , to return ModelMapper instance to SC.

Bean config xml file equivalent to : @Configuration : class level annotation

<bean> Tags --- @Bean : method level annotation(this method should ret bean instance to the caller : SC)

More Details here :

What is the difference between @Bean vs @Component ?

@Component n it's subtypes are Preferable for component scanning and automatic wiring.

When should you use @Bean?

Sometimes automatic configuration is not an option.

When?

Imagine that you want to wire components from 3rd-party libraries , eg : PasswordEncoder in SpringSecurity or ModelMapper (you don't have the source code so you can't annotate its classes with @Component), so automatic configuration is not possible.

The @Bean , method level annotation , which returns an object that spring should register as bean in application context.

The body of the method should create the instance n ret it to the caller (SC) , for it's auto management as a spring bean.

eg :

@Bean //equivalent to <bean id/> in xml file

```
public ModelMapper modelMapper() {
  ModelMapper modelMapper = new ModelMapper();
  modelMapper.getConfiguration().setMatchingStrategy(MatchingStrategies.STRICT) //only MATCHING
  prop names between src n dest will be transferred , during the mapping
  .setPropertyCondition(Conditions.isNotNull()); // only non null properties will be transferred from src -->
  dest , during the mapping
  return modelMapper;
}
```

1.3 Inject ModelMapper as a dependency in other spring beans
(@AutoWired)

1.4 Use ModelMapper's map method , to map Entity <----> DTO

```
public D map(T src,Class<D> class)
```

T : src type of object

D : type of detination.

It will create the instance of destination type class n set MATCHING props from the src object

eg : To map Employee entity ref --> emp resp dto

```
EmpRespDTO dto=mapper.map(empEntity,EmpRespDTO.class);
```